

Define a quantity from scalar and unit

```
>>> from astropy import units as unit
```

```
>>> B = 1.0 * unit.gauss
```

```
>>> B
```

```
<Quantity 1. G>
```

```
>>> B.si
```

```
<Quantity 0.0001 T>
```

```
>>> import numpy as np
```

Import constants and use them with scalars

```
>>> from axionbloch.constants import gamma_p
```

```
>>> from axionbloch.dependency import PI
```

Find the 90 degree pulse duration

```
>>> t90 = PI / 2 / (gamma_p * B)
```

```
>>> t90.to(unit.microsecond)
```

```
<Quantity 58.71648793 us>
```

Operation on an array of quantities with numpy

```
>>> tStamps = np.array([0, 1 / 3, 1]) * t90
```

```
>>> phases = gamma_p * B * tStamps
```

```
>>> np.sin(phases)
```

```
<Quantity [0. , 0.5, 1. ]>
```